Supplementary Information for:

Constructing the Nation: Identity and Historical Narratives in South Korean History Textbooks

Steven Denney* Aron van de Pol†

April 15, 2026

*Assistant Professor, Leiden University (s.c.denney@hum.leidenuniv.nl)
†PhD Candidate, Leiden University (a.m.van.de.pol@hum.leidenuniv.nl)

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1 Appendix A: Robustness checks for term discovery

Introduction

This appendix tests whether the 12 identity terms discovered by the LLM pipeline reflect genuine structure in the textbook corpus rather than artifacts of generative extraction. We compare the LLM output to two alternative topic-modeling approaches drawn from different methodological traditions: BERTopic, which clusters semantically similar text chunks in embedding space, and Latent Dirichlet Allocation (LDA), which models document-level bag-of-words co-occurrence. The goal is not exact agreement. It is to see whether the concepts selected by the LLM pipeline reappear when the corpus is organized by substantively different methods.

Methods

The baseline term list came from the three-model EXAONE pipeline described in the main text and Appendix D. Each model extracted identity-related nouns from sampled textbook windows using a Korean zero-shot prompt. Candidate terms were then validated by cosine similarity between the term embedding and the embedding of the source window. Surviving candidates were scored with a weighted composite of normalized frequency (40%) and semantic similarity (60%). Within each model, terms above the $\mu + 2\sigma$ threshold were retained. Final inclusion required agreement by at least two of the three models and an average weighted frequency of at least 300. This procedure produced the 12 terms used in the main analysis.

For the first robustness check, we ran BERTopic on the full corpus after splitting each textbook into paragraph-scale chunks targeting roughly 500 characters.[1] This yielded 29,568 chunks with a mean length of 433.34 characters. Chunks were embedded with `BM-K/KoSimCSE-`

[1] We segment at the paragraph scale because transformer-based embedding models truncate long inputs, and BERTopic assumes each input unit contains a reasonably coherent local topic.

`roberta-multitask`, the same Korean sentence-embedding model used in the main semantic analysis, then projected with UMAP and clustered with HDBSCAN. Topic representations were built with c-TF-IDF using a KiwiPiePy noun tokenizer. With a minimum topic size of 30, BERTopic identified 214 non-outlier topics and classified 8,804 chunks as outliers, an outlier rate of 29.8%. A term counted as recovered if it appeared among the top 30 representative words of at least one topic.

For the second robustness check, we ran LDA on the noun-extracted `processed_text` field, which provides a document-level bag-of-words representation of the 67 textbooks. After filtering rare and very common terms, the vocabulary contained 20,155 tokens. We estimated models for $k \in \{10, 15, 20, 25, 30\}$ and used c_v coherence with 20 passes to choose the preferred specification. The best model was $k = 20$ with $c_v = 0.5164$. As with BERTopic, a term counted as recovered if it appeared among the top 30 words of at least one topic. We also aggregated document-topic proportions by period as a face-validity check on whether the recovered topics displayed plausible historical patterning.

Method	**Unit**	**Input size**	**Topics**	**Key specification**	**Coverage**	**Terms missed**
BERTopic	Chunk-level semantic clustering	29,568 chunks from 67 textbooks	214	KoSimCSE embeddings; min topic size 30; 8,804 outliers (29.8%)	12/12 (100%)	None
LDA	Document-level bag-of-words	67 textbooks; 20,155-term vocab.	20	Best $k = 20$; $c_v = 0.5164$; 20 passes	7/12 (58.3%)	공동체, 국가, 역사, 영토, 해방

Table S1: Summary of the two robustness checks.

Results: term recovery

Table S1 summarizes the model-level comparison, and Table S2 reports recovery for each of the 12 LLM terms. BERTopic recovered all 12 terms, yielding 100.0% coverage. LDA recovered 7 of 12 terms, or 58.3%. The same 7 terms were recovered by both methods, and all 12 were recovered by at least one of them. No LLM-discovered term therefore failed both independent checks.

The terms recovered by both BERTopic and LDA are 국민 (citizen/national), 독립 (independence), 민족 (nation/ethnos), 민주주의 (democracy), 사회 (society), 정치 (politics), and 통일 (unification). These are the most robustly attested identity concepts in the corpus by this criterion. The remaining five terms, 공동체 (community), 국가 (state), 역사 (history), 영토 (territory), and 해방 (liberation), are recovered by BERTopic but not by LDA. The difference is informative. It suggests that the LLM pipeline captures concepts that are visible in local semantic structure even when document-level topic modeling cannot isolate them cleanly.

LLM term	BERTopic	LDA	Both
community (*kongdongch'e*, 공동체)	Yes (25)	No	No
state (*kukka*, 국가)	Yes (22)	No	No
citizen/national (*kungmin*, 국민)	Yes (1)	Yes (3)	Yes
independence (*tongnip*, 독립)	Yes (1)	Yes (3)	Yes
nation/ethnos (*minjok*, 민족)	Yes (2)	Yes (1)	Yes
democracy (*minjujuŭi*, 민주주의)	Yes (3)	Yes (27)	Yes
society (*sahoe*, 사회)	Yes (11)	Yes (1)	Yes
history (*yŏksa*, 역사)	Yes (1)	No	No
territory (*yŏngt'o*, 영토)	Yes (4)	No	No
politics (*chŏngch'i*, 정치)	Yes (7)	Yes (3)	Yes
unification (*t'ongil*, 통일)	Yes (3)	Yes (10)	Yes
liberation (*haebang*, 해방)	Yes (9)	No	No

Table S2: Recovery of LLM-discovered identity terms in BERTopic and LDA. Parentheses report the best rank achieved across topics.

Results: explaining the LDA gap

The five LDA misses are interpretable. Two of them, 국가 (state) and 역사 (history), are highly ubiquitous across the corpus. 국가 appears in 64 of 67 textbooks, and 역사 appears in all 67. Because LDA privileges words that distinguish one topic from another, broadly distributed terms often fail to rank highly in any single topic even when they are substantively central. 공동체 reflects the opposite problem. With only 236 corpus occurrences, it is too sparse to surface consistently among the top-ranked words of a document-level topic model. 영토, which appears in 60 textbooks with 1,181 occurrences, and 해방, which appears in 46 textbooks with 507 occurrences, fall between these two extremes.

BERTopic is less constrained by these limits. It operates on 29,568 local chunks rather

than 67 full documents, which provides much finer statistical resolution, and it clusters by semantic affinity rather than token co-occurrence alone. That helps explain why BERTopic recovers all five terms that LDA misses. The LDA gap is therefore informative, not disconfirming. It shows that the LLM pipeline captures both topic-discriminative concepts and broader background identity vocabulary that document-level bag-of-words modeling cannot easily isolate.

Results: period patterns

Period distributions provide a limited but useful face-validity check. In BERTopic, the clearest unification topic, defined by terms such as 남북, 남북한, 통일, 회담, and 평화, is most prevalent in the Democratic period, where it accounts for 1.84% of chunks, compared with 0.29% in the Authoritarian period. The clearest democracy topic, centered on 대통령, 민주, 선거, and 헌법, also peaks in the Democratic period at 2.78% of chunks. LDA's period patterns are much less interpretable because identity terms are dispersed across broad mixed topics, but they do not contradict the broader conclusion that the recovered terms align with recognizable historical themes.

Discussion

The robustness checks support the main analysis without requiring stronger claims than the data warrant. BERTopic independently recovers all 12 LLM terms, and LDA recovers 7. The 7 terms found by all three procedures, LLM consensus, BERTopic, and LDA, are the most stable identity concepts in the corpus by a conservative standard. The five BERTopic-only terms are also substantively plausible. They include highly ubiquitous concepts such as 국가 and 역사, and low-frequency concepts such as 공동체, which are exactly the kinds of terms document-level LDA is least likely to isolate.

The three approaches answer slightly different questions. Topic models ask which latent themes organize the corpus. The LLM pipeline asks which concepts in the corpus are relevant

to national identity. For the present study, that second question is the one we need answered directly. BERTopic and LDA therefore function as independent checks rather than substitutes. Neither would have produced the same twelve-term list on its own. Taken together, they show that the LLM-discovered terms are grounded in recoverable textual structure rather than being artifacts of the extraction process.

2 Appendix B: Corpus inventory

Table S3 lists all 67 textbooks in the merged corpus, grouped by the five-period classification used in the main text. The table follows the metadata stored in `00_data/nikh_corpus_merged.csv`. We include each textbook's stable identifier (`book_id`) so the printed inventory can be matched directly to the replication files. Some early and OCR-derived items have incomplete bibliographic metadata. In those cases, the table reproduces the fields available in the merged corpus rather than inferring missing values.

Period	**book_id**	**Title**	**Year**	**Publisher**	**Lv.**
Late Chosŏn	ta_p111r	국조사		개인출판	
	ta_p101r	대동청사 1책		개인출판	
	ta_p102r	대동청사 2책		개인출판	
	ta_p21r	조선역대사략 권1	1895	학부 편집국	Pub.
	ta_p22r	조선역대사략 권2	1895	학부 편집국	Pub.
	ta_p23r	조선역대사략 권3	1895	학부 편집국	Pub.
	ta_p11r	조선역사 상	1895	학부 편집국	Pub.
	ta_p12r	조선역사 중	1895	학부 편집국	Pub.
	ta_p13r	조선역사 하	1895	학부 편집국	Pub.
	ta_p71r	보통교과 동국역사(1권)	1899		E
	ta_p72r	보통교과 동국역사(2권)	1899		E
	ta_p81r	동국사략 상	1906		
	ta_p82r	동국사략 하	1906		
	ta_p91r	초등대한역사	1908		E
Colonial	ta_p41r	심상소학국사보충교재 교수참고서 - 1		문부성	
	ta_p42r	심상소학국사보충교재 교수참고서 - 2		문부성	
	ta_p31r	심상소학국사보충아동용 - 1		문부성	
	ta_p32r	심상소학국사보충아동용 - 2		문부성	
Postwar	ta_p51r	국사교본		진단학회	
	ta_p61r	우리나라의 생활		진단학회	
	ta_h11	고등학교 국사 1차	1954	문교부	H
	ta_m11	중학교 국사 1차	1954	문교부	M
	ta_e11	초등학교 사회생활 6-1(1차)	1954	문교부	E
	ta_h21	고등학교 국사 2차	1963	문교부	H
	ta_m21	중학교 국사 2차	1963	문교부	M
	ta_e21	초등학교 사회 6-1(2차)	1963	문교부	E

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Table S3 continued

Period	book_id	Title	Year	Publisher	Lv.
Authoritarian	ta_h31	고등학교 국사 3차	1973	문교부	H
	ta_h32	고등학교 세계사(3차)	1973	문교부	H
	ta_m31	중학교 국사 3차	1973	문교부	M
	ta_e31	초등학교 국사 5(3차)	1973	문교부	E
	ta_e32	초등학교 국사 6(3차)	1973	문교부	E
	ta_h41	고등학교 국사 4차(상)	1981	문교부	H
	ta_h42	고등학교 국사 4차(하)	1981	문교부	H
	ta_m41	중학교 국사 4차(상)	1981	문교부	M
	ta_m42	중학교 국사 4차(하)	1981	문교부	M
	ta_e41	초등학교 사회 5-2(4차)	1981	문교부	E
	ta_e42	초등학교 사회 6-2(4차)	1981	문교부	E
	ta_h51	고등학교 국사 5차(상)	1987	교육부	H
	ta_h52	고등학교 국사 5차(하)	1987	교육부	H
	ta_e51	역대 국사교과서	1987	교육부	E
	ta_m51	중학교 국사 5차(상)	1987	교육부	M
	ta_m52	중학교 국사 5차(하)	1987	교육부	M
	ta_e52	초등학교 사회 6-1(5차)	1987	교육부	E
	H-18(1,90)1	우리는 민족 중흥의 역사적 사명을 띠고 이 땅에 태어났	1990	문교부	H
	H-19(1,90)1	국사	1991	문교부	M
Democratic	ta_h61	고등학교 국사 6차(상)	1995	교육부	H
	ta_h62	고등학교 국사 6차(하)	1995	교육부	H
	ta_m61	중학교 국사 6차(상)	1995	교육부	M
	ta_m62	중학교 국사 6차(하)	1995	교육부	M
	ta_e61	초등학교 사회 6-1(6차)	1995	교육부	E
	H-21(1,2001)1	국사 편찬 위원회	2001	문교부	M
	ta_h71	고등학교 국사 7차	2002	교육인적자원부	H
	H-2(1,2002)	국사 편찬 위원회	2002	교육인적자원부	H
	ta_m71	중학교 국사 7차	2002	교육인적자원부	M
	ta_e71	초등학교 사회 6-1(7차)	2002	교육인적자원부	E
	H-6(1,2007)	한국근현대사	2002	중앙교육진흥연구소	H
	H-7(1,2007)	한국근현대사	2002	법문사	H
	H-9(5,2007)	한국근현대사	2002	대한교과서(주)	H
	H-20(1,2001)1	국사 편찬 위원회	2004	문교부	H
	H-2(2,2006)	국사 편찬 위원회	2006	교육인적자원부	H
	H-1(1,2006)	국사편찬위원회 국정도서편찬위원회	2006	교육인적자원부	M
	H-5(1,2007)	한국근현대사	2007	금성출판사	H

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Table S3 continued

Period	book_id	Title	Year	Publisher	Lv.
	H-8(1,2007)	한국근현대사	2007	두산	H
	H-10(4,2007)	한국근현대사	2008	천재교육	H
	H-39(1,2002)	국사 편찬 위원회	2009	교육인적자원부	M
	H-57(1,2016)	한국사	2013	지학사	H
	H-54(2,2016)	한국사	2016	금성출판사	H

Table S3: Full corpus inventory by period. N = 67 textbooks spanning 1895–2016. Level abbreviations: E = Elementary, M = Middle school, H = High school, Pub. = public-school audience in the original metadata. Blank cells indicate unavailable metadata in the merged corpus.

3 Appendix C: Text processing pipeline

This appendix documents the OCR, cleaning, and morphological processing steps applied to the corpus before analysis.

C.1 OCR for GEI textbooks

Sixteen textbooks from the Georg Eckert Institute (GEI) were available only as scanned PDFs. We processed these books with DeepSeek-OCR, a 3-billion-parameter vision-language model run in BF16 precision. The model converts page images into structured markdown while preserving broad document layout, including tables, lists, and heading structure. Pages were processed at 200–300 DPI on GPU infrastructure (CUDA 12.4) with flash attention. Processing time was approximately 10–30 seconds per page.

C.2 Artifact removal

Raw OCR output contained residual HTML tags, repeated punctuation from decorative borders, stray symbols, and LaTeX remnants. A post-processing script removed these artifacts and reduced total character count across the 16 GEI textbooks by 23.3%. The cleaning script is included in the replication archive as `03_preprocessing/clean_ocr_artifacts.py`.

C.3 Morphological analysis

Because Korean is agglutinative, a single lexical concept can appear with many particles and inflectional endings. To reduce that source of noise in token-based analysis, we used `kiwipiepy` (Kiwi) to build a noun-focused text field stored as `processed_text`.

The processing pipeline had seven steps:

1. basic cleaning of URLs, e-mail addresses, and special characters;

2. morphological analysis with Kiwi;

3. part-of-speech filtering to retain only common nouns (NNG) and proper nouns (NNP);
4. compound-noun merging to preserve names and technical expressions;
5. stopword removal using a 679-item Korean stopword list;
6. length filtering to drop single-syllable tokens unless recognized as proper nouns, with tokens restricted to Hangul and Hanja ranges; and
7. number removal.

Across the full corpus, character count fell from 12,988,638 to 4,746,884, a reduction of 63.5%. The processed corpus contains 67 documents with a mean of 22,406 tokens per document. No documents were dropped.

The unfiltered `full_text` field remains available for analyses that require clause structure or verbal morphology. The `processed_text` field is used for exact term counts and other token-based analyses. The sentence-embedding analyses in the main text use `full_text`, because the embedding model requires natural sentence structure.

4 Appendix D: Term discovery pipeline

The main text outlines the term discovery pipeline. This appendix gives the full specification.

D.1 Extraction prompt

Each text window was presented to the model with the following Korean-language zero-shot instruction:

> 이 텍스트에서 '우리는 누구인가', '어떤 사람이 한국인인가', '한국이란 무엇인가'와 같은 질문에 답하려는 모든 관련 핵심 용어와 표현을 추출하라.

The prompt then specified seven thematic domains: collective belonging, community consciousness, nation and ethnicity, state and territory, unification/independence/liberation, democracy and political systems, and historical legitimacy and values. The model was instructed to return only a JSON array of concrete nouns and to exclude meta-words such as 용어, 항목, and 예시, as well as any term not present in the source text.

D.2 Adaptive window sampling

For each of the 67 textbooks, we sampled text windows of 4,000 characters. The number of windows per document scaled with document length, with a minimum of 3 and a maximum of 50. Sampling always anchored the beginning, middle, and end of the document before filling the remaining slots with random positions. This guarantees coverage of opening, core, and closing sections regardless of book length.

D.3 Root word extraction

Raw LLM outputs often contained inflected forms, compounds, or multi-morpheme phrases. KiwiPiePy tokenized each extracted expression and retained the longest noun token tagged

as NNG or NNP. For example, “민족주의적” was reduced to “민족주의,” and “국가 건설” to “국가.” This step aggregates morphological variants before scoring.

D.4 Scoring formula

After semantic-similarity validation and corpus-frequency filtering, surviving terms were scored by period with the following weighted composite:

$$\text{score} = 0.4 \times \hat{f} + 0.6 \times \hat{s} \tag{1}$$

Here $\hat{f}$ is min-max normalized frequency in the period and $\hat{s}$ is the min-max normalized cosine similarity between the term embedding and its source-window embedding, computed with `paraphrase-multilingual-MiniLM-L12-v2`. The heavier weight on similarity prevents frequent but weakly relevant nouns, such as 사람 or 시대, from dominating the candidate list.

Scores were then aggregated to the root-word level across periods through weighted frequency (score $\times$ document frequency), yielding a single `total_weighted_frequency` value per root word and model. Terms above the $\mu + 2\sigma$ threshold on this measure were retained as each model’s candidate list.

D.5 Cross-model consensus

The three per-model candidate lists were merged. A term entered the final list only if:

- it was selected by at least 2 of the 3 models, **and**
- its average `total_weighted_frequency` across models exceeded 300.

This produced the 12 identity terms used in all downstream analyses (Table S4).

Term	n_models	Avg. weighted freq.
community (*kongdongch'e*, 공동체)	2	318
state (*kukka*, 국가)	3	1,247
citizen / national (*kungmin*, 국민)	3	892
independence (*tongnip*, 독립)	3	1,104
nation / ethnos (*minjok*, 민족)	3	1,589
democracy (*minjujuŭi*, 민주주의)	3	451
society (*sahoe*, 사회)	3	1,356
history (*yŏksa*, 역사)	3	987
territory (*yŏngt'o*, 영토)	2	412
politics (*chŏngch'i*, 정치)	3	1,178
unification (*t'ongil*, 통일)	3	876
liberation (*haebang*, 해방)	2	334

Table S4: Final identity terms with cross-model agreement. *n_models* = number of EXAONE variants that independently selected the term.

Note on 정체성 (*identity*). A thirteenth term, 정체성, was included in early analyses through an always-include rule based on its theoretical relevance. Subsequent inspection showed that in this corpus the term overwhelmingly refers to 停滯性 (stagnation), a historiographical concept used to counter Japanese colonial claims about Korean historical backwardness, rather than 正體性 (identity). It was therefore excluded from the final 12-term list and from all results reported in the main text. Appendix F discusses the issue further.

5 Appendix E: Term frequencies by education level

Table S5 disaggregates the period-level frequency analysis by school level: Elementary, Middle School, High School, and Pre-modern for texts that predate the modern school system. Frequencies are reported per 1,000 tokens in `processed_text` and averaged across textbooks within each period-level cell. Standard deviations appear in parentheses. Cells with 0.00 indicate that the term was absent from all textbooks in that cell.

Term	Level	Period				
		Late Chosŏn	Colonial	Postwar	Authoritarian	Democratic
nation (*minjok*, 민족)	Elem.	0.00	—	8.25 (6.21)	12.40 (11.09)	13.22 (4.84)
	Middle	—	—	6.50 (0.33)	11.74 (7.12)	10.89 (3.29)
	High	—	—	5.65 (3.37)	6.79 (4.14)	7.62 (3.93)
	Pre-mod.	0.05 (0.17)	0.00	4.75 (3.16)	—	—
state (*kukka*, 국가)	Elem.	0.84 (0.53)	—	6.95 (3.28)	4.88 (4.18)	2.23 (0.11)
	Middle	—	—	7.19 (1.60)	7.21 (2.02)	4.80 (1.48)
	High	—	—	6.14 (1.67)	5.95 (1.89)	4.31 (1.51)
	Pre-mod.	0.47 (0.56)	1.17 (0.58)	3.15 (0.89)	—	—
citizens (*kungmin*, 국민)	Elem.	0.00	—	14.78 (1.03)	7.36 (7.48)	3.06 (1.12)
	Middle	—	—	3.20 (0.04)	2.87 (1.75)	2.26 (1.37)
	High	—	—	1.18 (1.48)	2.21 (1.25)	2.29 (1.36)
	Pre-mod.	0.05 (0.09)	0.17 (0.20)	1.85 (1.73)	—	—
democracy (*minjujuŭi*, 민주주의)	Elem.	0.00	—	9.10 (2.39)	2.33 (4.22)	0.39 (0.23)
	Middle	—	—	0.67 (0.62)	0.13 (0.12)	0.45 (0.37)
	High	—	—	0.09 (0.03)	0.38 (0.38)	0.76 (0.51)
	Pre-mod.	0.00	0.00	0.03 (0.04)	—	—
independence (*tongnip*, 독립)	Elem.	0.38 (0.66)	—	6.21 (0.27)	4.78 (4.02)	11.34 (6.70)
	Middle	—	—	4.76 (0.12)	6.29 (6.79)	5.47 (4.33)
	High	—	—	2.55 (2.84)	3.94 (4.33)	5.39 (3.64)
	Pre-mod.	0.14 (0.30)	0.34 (0.39)	1.40 (0.47)	—	—
unification (*t'ongil*, 통일)	Elem.	0.45 (0.61)	—	5.36 (1.15)	3.78 (4.03)	3.39 (1.27)
	Middle	—	—	4.36 (0.67)	4.91 (2.42)	4.38 (1.24)
	High	—	—	2.98 (0.47)	2.98 (1.27)	3.03 (0.66)
	Pre-mod.	0.36 (0.40)	2.63 (3.04)	2.04 (1.10)	—	—
society (*sahoe*, 사회)	Elem.	0.00	—	7.86 (1.11)	5.84 (3.27)	3.39 (1.59)
	Middle	—	—	5.82 (0.95)	10.23 (1.53)	6.90 (1.98)
	High	—	—	8.06 (0.73)	10.67 (3.92)	7.88 (2.94)
	Pre-mod.	0.03 (0.07)	0.11 (0.13)	2.48 (0.39)	—	—
politics (*chŏngch'i*, 정치)	Elem.	0.62 (0.70)	—	15.10 (6.29)	6.92 (7.61)	3.54 (2.77)
	Middle	—	—	10.97 (1.96)	7.63 (0.75)	8.05 (0.87)
	High	—	—	5.35 (1.33)	7.35 (1.45)	4.41 (1.79)
	Pre-mod.	1.26 (1.59)	3.17 (3.68)	4.89 (0.84)	—	—
history (*yŏksa*, 역사)	Elem.	1.10 (0.16)	—	3.92 (1.01)	5.18 (3.41)	4.04 (2.13)
	Middle	—	—	3.79 (1.36)	3.91 (0.63)	4.71 (2.63)
	High	—	—	1.88 (0.26)	2.13 (0.33)	3.08 (1.00)
	Pre-mod.	0.47 (0.43)	0.62 (0.58)	1.62 (0.03)	—	—
territory (*yŏngt'o*, 영토)	Elem.	0.49 (0.63)	—	0.81 (0.05)	0.75 (0.77)	1.27 (1.00)
	Middle	—	—	1.60 (0.24)	1.08 (0.46)	1.27 (0.46)
	High	—	—	0.39 (0.02)	0.85 (0.59)	0.58 (0.36)
	Pre-mod.	0.30 (0.59)	2.83 (2.70)	0.99 (0.42)	—	—
liberation (*haebang*, 해방)	Elem.	0.05 (0.08)	—	1.24 (0.87)	0.35 (0.66)	0.00

Continued on next page

Table S5 continued

Term	**Level**	**Late Chosŏn**	**Colonial**	**Postwar**	**Authoritarian**	**Democratic**
	Middle	—	—	0.84 (0.05)	0.20 (0.12)	0.36 (0.05)
	High	—	—	0.45 (0.35)	0.27 (0.26)	0.35 (0.20)
	Pre-mod.	0.10 (0.20)	0.00	0.57 (0.28)	—	—
community (*kongdongch'e*, 공동체)	Elem.	0.00	—	0.00	0.00	0.00
	Middle	—	—	0.04 (0.06)	0.03 (0.05)	0.09 (0.05)
	High	—	—	0.23 (0.13)	0.20 (0.10)	0.24 (0.10)
	Pre-mod.	0.00	0.00	0.00	—	—

Table S5: Mean frequency per 1,000 tokens by term, period, and education level. Standard deviations are in parentheses. "—" indicates that no textbooks at that level are present in the period. Elem. = Elementary; Pre-mod. = Pre-modern texts predating the modern school system.

The level breakdown clarifies several patterns that are muted in the period-level aggregates. *Kungmin* is sharply elevated in postwar elementary textbooks (14.78 per 1,000 tokens), more than four times its frequency in postwar high school texts (1.18), which suggests that early republican civic socialization was concentrated in primary education. *Minjujuŭi* follows a similar pattern: 9.10 in postwar elementary texts and only 0.39 in democratic-era elementary texts. As argued in the main text, democracy shifts from an explicit pedagogical aspiration to a more taken-for-granted background condition. *Minjok* is consistently more frequent in elementary and middle school textbooks than in high school texts across all post-liberation periods, which suggests that ethnic-national framing was strongest in materials for younger students. *Yŏngt'o* peaks in colonial-era pre-modern texts at 2.83, the highest value for any term before liberation, consistent with the territorial emphasis of imperial historiography.

Appendix F: Full-timeline vs. postwar-only comparison

The main text focuses on post-liberation transitions (Postwar → Authoritarian → Democratic). This appendix reports the full five-period analysis that motivated that restriction and compares the full-timeline and postwar-only logistic deltas.

F.1 Why restrict to post-liberation periods?

The full-timeline analysis included the Late Chosŏn and Colonial periods alongside the three post-liberation periods. The Colonial → Postwar transition produced mean logistic deltas of 1.97, roughly 2.3 times larger than any later transition. That scale difference reflects a qualitatively different kind of change: the replacement of Japanese imperial historiography with post-liberation nationalist education, not a shift within a continuous discourse tradition.

Including the early periods also changes the scaling baseline. The logistic axis and the StandardScaler normalization are estimated over all loaded data. Nineteenth-century and colonial texts shift that baseline and can inflate apparent authoritarian-to-democratic differences for terms that are absent or marginal before liberation, such as 민주주의 and 공동체. The postwar-only analysis therefore isolates regime-type effects within the continuous post-liberation discourse more cleanly.

F.2 Logistic delta comparison

Table S6 compares the authoritarian-to-democratic logistic deltas from the full five-period run and the postwar-only run. Terms are sorted by the magnitude of the difference.

The comparison sorts the terms into three groups.

Robust (9 terms). The authoritarian-to-democratic shift remains positive in both analyses. These terms show a genuine regime-linked effect: their semantic contexts change when political institutions change, regardless of whether the early periods are included. The magnitude is smaller in the postwar-only run, which is what we would expect once the

Term	Δ_{full}	Δ_{pw}	Diff.	Label
citizen (*kungmin*, 국민)	+0.449	+0.378	−0.071	Robust
independence (*tongnip*, 독립)	+0.389	+0.313	−0.076	Robust
nation (*minjok*, 민족)	+0.402	+0.280	−0.122	Robust
community (*kongdongch'e*, 공동체)	+0.492	+0.280	−0.212	Robust
state (*kukka*, 국가)	+0.328	+0.278	−0.050	Robust
society (*sahoe*, 사회)	+0.374	+0.254	−0.119	Robust
politics (*chŏngch'i*, 정치)	+0.335	+0.213	−0.122	Robust
history (*yŏksa*, 역사)	+0.449	+0.167	−0.282	Robust
unification (*t'ongil*, 통일)	+0.476	+0.114	−0.363	Robust
liberation (*haebang*, 해방)	+0.416	+0.014	−0.402	Flat
territory (*yŏngt'o*, 영토)	+0.339	−0.024	−0.363	Flat
democracy (*minjujuŭi*, 민주주의)	+0.496	−0.073	−0.569	Reversed
identity/stagnation (*chŏngch'esŏng*, 정체성)	+0.493	−0.277	−0.770	Reversed

Table S6: Authoritarian-to-democratic logistic deltas: full timeline vs. postwar only. Δ_{full} = delta from the five-period analysis. Δ_{pw} = delta from the postwar-only analysis. Difference = $\Delta_{\text{pw}} - \Delta_{\text{full}}$.

pre-liberation baseline is removed, but the direction is stable.

Flat (2 terms). *Haebang* (liberation) and *yŏngt'o* (territory) show sizeable positive deltas in the full run but near-zero deltas in the postwar-only run. Their full-run signal is driven mainly by the long historical baseline rather than by authoritarian-to-democratic change. In the post-liberation sequence, liberation discourse remains fairly stable, while territory continues to revolve around longstanding historical and geopolitical themes.

Reversed (2 terms). *Minjujuŭi* (democracy) and *chŏngch'esŏng* (정체성) show positive deltas in the full run but negative deltas in the postwar-only run. In both cases, the full-run result is misleading.

For 민주주의, the reversal points to a substantive result rather than a defect in the method. Authoritarian and democratic textbooks often use democracy in closely overlapping semantic contexts. The authoritarian regime did not remove democratic vocabulary from the textbooks. It appropriated it. The logistic classifier therefore has little leverage for distinguishing the two periods on this term once the early baseline is removed.

For 정체성, the reversal is a homophone artifact. In this corpus, 정체성 overwhelm-

ingly refers to 停滯性 (stagnation), a historiographical concept used in debates over colonial interpretations of Korean history, rather than 正體性 (identity). Both authoritarian and democratic textbooks use the term in that sense. This is why the term was excluded from the main analysis.

F.3 Implications

The comparison shows why period-restricted robustness checks matter in long-horizon semantic analysis. When the baseline includes qualitatively different discourse traditions, the resulting deltas can mix foundational rupture with the within-tradition adjustments we actually want to study. All findings in the main text therefore use the postwar-only analysis. The full-timeline results remain useful as historical context, especially for demonstrating that the largest rupture in the corpus comes at liberation rather than at later regime transitions.

6 Appendix G: Exemplar sentences

For each identity term discussed in the main analysis, we selected sentences from the extreme poles of all three semantic axes: contrastive variance (cPCA), discriminative (logistic regression), and alignment residual (Procrustes). These are the sentences whose embeddings (KoSimCSE-roberta, 768 dimensions) project most strongly in each direction along each axis. Examples already quoted in the main text are not repeated here.

The three axes capture different geometric properties of semantic change and sometimes place the same sentence at opposite poles. We note these reversals where they occur. Source textbook years are given in parentheses; the full exemplar set with axis coordinates and book identifiers is in the replication archive.

G.1 국민 (*kungmin*, citizen)

Postwar → Authoritarian. *Contrastive variance.* At the Postwar-direction pole, *kungmin* appears in the language of traditional governance:

> 국왕의 정책 결정이 잘못되었을 때에는 사간원이 간언을 올릴 수 있었고, 국민들에게도 정치적 의견을 제시할 수 있도록 상소의 길이 열려 있었다.
>
> "*When royal policy decisions were mistaken, the Saganwŏn could submit remonstrance, and a path for petitions was open so that citizens could present political opinions as well.*" (1981)

At the Authoritarian-direction pole, the National Education Charter dominates. All three axes place the Charter at this pole.

> 국민 교육 헌장
>
> "*National Education Charter.*" (1981)

Discriminative. The logistic axis separates factional politics from civic improvement. At the Postwar pole:

> 당 이외에는 아무 것도 보이지 않아 임진왜란 중에 쫓겨 가면서도 이 싸움이요, 남한산성에 포위를 당한 중에도 이 분열이니 그들의 눈에는 국가의 장래와 국민의 생활을 돌아볼 여가가 없었다.

> "*They could see nothing but their faction. Even while fleeing during the Imjin War, it was still this fighting. Even while besieged at Namhansansŏng, it was still this division. They had no time to consider the nation's future or the citizens' livelihood.*"

At the Authoritarian pole:

> 한편, 일부 학자들 사이에는 학문 연구가 국민 생활을 향상시키는 데 기여해야 한다는 생각을 가지고 새로운 학문적 분위기를 만들어 나가기도 하였다.
>
> "*Meanwhile, some scholars held the view that academic research should contribute to improving citizens' lives, and worked to create a new scholarly atmosphere.*" (1981)

Alignment residual. Procrustes agrees with cPCA: the Charter at one pole, and at the other, a democratic-era framing of citizens' rights:

> 오늘날에는 국민의 자유와 권리가 이전보다 더 잘 보장되고, 국민의 여러 가지 주장이 자유롭게 발표되고 있으며, 각 정당들의 활동도 활발해졌다.
>
> "*Today, citizens' freedoms and rights are better guaranteed than before, citizens' various opinions are freely expressed, and political party activities have become vigorous.*" (1987)

Authoritarian → Democratic. *Contrastive variance.* The negative pole (Authoritarian direction) is dominated by democratic-era sentences about economic hardship:

> 외환 위기로 인한 국민들의 고통은 엄청난 것이었다.
>
> "*The suffering of the citizens caused by the foreign exchange crisis was enormous.*" (2008)

The positive pole sentences are textbook index tables rather than narrative prose, which is itself diagnostic: *kungmin* in democratic textbooks appears so frequently in organizational apparatus that its highest-scoring pole projections are structural rather than discursive. *Discriminative.* The logistic axis captures a different contrast. At the Authoritarian pole, *kungmin* appears in European diplomatic history:

> 그러나, 다시 오스트리아가 알바니아를 세움으로써 뜻을 이루지 못하게 되자, 오스트리아에 대한 국민 감정이 극도로 악화되었다.
>
> "*However, when Austria established Albania and thus thwarted [Serbia's] ambitions, national sentiment toward Austria deteriorated to the extreme.*" (1973)

At the Democratic pole, it enters environmental-policy language:

> 소득 수준뿐만 아니라, 국민 개개인의 삶의 질이 세계화하고 일류화하는 환경 모범 국가를 목표로 하고 있다.
>
> "*The goal is an environmentally exemplary state in which both income levels and the quality of life of individual citizens become globalized and world-class.*" (2002)

Alignment residual. Procrustes reverses cPCA: the 외환 위기 sentence appears at the *positive* pole, and the index tables at the negative. The two axes identify the same contrast but orient it differently in the embedding space.

G.2 독립 (*tongnip*, independence)

Postwar → Authoritarian. *Contrastive variance.* The Postwar-direction pole frames independence as armed struggle:

> 이 밖에도 우리의 독립군이 여러 곳에서 일본군을 괴롭혔다.
>
> "*Besides this, our independence army harassed the Japanese military in many places.*" (1981)

The top-ranked Authoritarian-direction sentence is quoted in the body of the paper. The remaining pole is dominated by short institutional labels for the Independence Club and the *Tongnip Sinmun*:

> 최초의 본격적인 민간지는 독립 신문이었다.
>
> "*The first fully-fledged private newspaper was the Independent.*" (1987)

The cPCA axis separates armed military resistance at the Postwar pole from civic-institutional reform at the Authoritarian pole.

Discriminative. The logistic axis separates a Postwar chapter heading, "*Conservative and Independence factions,*" from architectural and cultural landmarks associated with the independence movement:

> 독립문과 덕수궁 안의 석조전, 종현 성당(명동 성당) 등 근대식 건물도 세워졌다.
>
> "*Modern buildings such as the Independence Gate, the Sŏkchojŏn Hall inside Tŏksugung Palace, and the Chonghyŏn Cathedral (Myŏngdong Cathedral) were also constructed.*" (1973)

Alignment residual. Procrustes reverses the cPCA poles. At the positive pole, where cPCA placed institutional reform, Procrustes places armed resistance:

> 청산리 대첩(민족 기록화) 김좌진이 지휘하는 독립군이 일본군과 혈전을 벌여 대승리를 거두었다.
>
> "*Battle of Ch'ŏngsanri* (*national historical painting*): *The independence army commanded by Kim Chwa-jin fought a bloody battle against the Japanese military and achieved a great victory.*" (1981)

This reversal means that the directions of maximum variance difference and maximum alignment residual are oriented differently. The cPCA axis separates the range of contexts (armed struggle versus institutional reform), while the Procrustes axis captures a structural relocation that runs in the opposite direction.

Authoritarian → Democratic. *Contrastive variance.* At the negative pole (Authoritarian direction), democratic textbooks generalize independence into a global frame:

> 힘이 약한 세계 여러 민족은 스스로 힘을 모아 독립을 얻었다.
>
> "*The various weak nations of the world gathered their own strength and achieved independence.*" (1995)

Discriminative. The logistic axis captures an entirely different sense of *tongnip*: not national independence but economic independence.

> 수공업자들의 독립 현상은 주로 놋그릇, 농기구, 모자, 장도 제조 분야에서 두드러졌다.
>
> "*The independence [i.e., independent operation] of artisans was especially prominent in the manufacture of brass vessels, agricultural tools, hats, and ornamental knives.*" (1987)

The logistic classifier treats this non-political sense as maximally characteristic of the Authoritarian period, which suggests that authoritarian textbooks more often used *tongnip* in descriptive economic contexts while democratic textbooks reserved the word for political independence.

G.3 민족 (*minjok*, nation)

Postwar → Authoritarian. *Contrastive variance.* Both top-ranked cPCA pole sentences are quoted in the body of the paper. The second-ranked Authoritarian-direction pole continues the colonial-grievance register:

> 그러나, 1940년대에 들어서는 민족성 말살 정책 때문에 이러한 활동마저 중단되어, 문화의 암흑기로 접어들게 되었다.
>
> "*However, entering the 1940s, even these activities were halted because of the policy to eradicate national identity, ushering in a dark age of culture.*" (1973)

Discriminative. The logistic axis captures an economic dimension absent from cPCA. At the Authoritarian pole:

> 그리고 각종의 근대적 회사와 공장, 은행 등을 설립하여 민족 자본에 의한 근대적 경제 체제를 이룩하려는 노력을 기울였다.
>
> "*And [they] made efforts to establish a modern economic system based on national capital by founding various modern companies, factories, and banks.*" (1987)

Alignment residual. The Procrustes negative pole captures the national division narrative:

> 군사적으로 그어진 38선이 정치적 분할선으로 굳어짐으로써 생겨난 우리 나라의 남북 분단은 우리 민족에게 커다란 시련을 안겨 주었다.
>
> "*The North-South division of our country, which arose when the militarily drawn 38th parallel hardened into a political dividing line, imposed a great ordeal on our nation.*" (1973)

At the positive pole, Procrustes places the growth of modern national consciousness:

> 근대 의식이 자라면서 교육열이 높아지고 민족 의식이 성장함에 따라, 국어에 대한 관심이 새로와지고 근대적인 역사 의식이 일어나기 시작하였다.
>
> "*As modern consciousness grew, educational fervor rose, and national consciousness developed, interest in the national language was renewed and a modern historical consciousness began to emerge.*" (1981)

Authoritarian → Democratic. The top cPCA negative pole sentence is quoted in the body of the paper. cPCA and Procrustes agree on this transition, placing the same sentences at the same poles. The cPCA positive pole (Democratic direction) invokes the colonial-era nationalist historiographic tradition:

> 민족사의 수난, 일제 식민 통치
>
> "*The ordeals of national history, Japanese colonial rule.*" (2007)

Discriminative. The logistic axis identifies a distinctive sentence at the Authoritarian pole that the variance axes miss:

> 그리하여, 1961년 5월 16일, 박정희 장군을 중심으로 한 군인들은, 나라와 민족을 혼란과 위기에서 구하기 위하여 혁명을 일으켰다.
>
> "*Thus, on May 16, 1961, soldiers centered on General Park Chung-hee launched a revolution to save the country and the nation from chaos and crisis.*" (1973)

This sentence does not appear at the extremes of either variance axis, yet the logistic classifier identifies it as maximally characteristic of the Authoritarian period. The framing of the 1961 coup as a national salvation act, described without qualification in a textbook produced under the subsequent curriculum, is precisely the kind of period-specific discourse that the discriminative axis is designed to capture.

G.4 해방 (*haebang*, liberation)

The contrastive axes for *haebang* do not separate different political framings of national liberation. Instead, they track the word's polysemy, separating 1945 liberation from colonial rule, premodern slave emancipation, and social-movement liberation. This polysemy structure is what drives the term's exceptionally large Procrustes shift during democratization, discussed in the main analysis.

Postwar → Authoritarian. *Contrastive variance.* At the negative pole (Postwar direction), *haebang* means national liberation from colonial rule, paired with the trauma of subsequent division:

> 1945년 해방 후에도 국토의 남북 분단과 민주주의 대 공산주의의 싸움이라는 세계적인 현실을 무릅쓰지 않을 수 없었다.
>
> "*Even after liberation in 1945, [Korea] could not escape the global reality of the North-South division of the homeland and the struggle between democracy and communism.*" (1973)

At the positive pole (Authoritarian direction), the word refers to premodern slave emancipation:

> 그는 먼저 노비 안검법을 실시하여 호족들이 불법으로 소유하고 있는 많은 노비를 해방시켰다.
>
> "*He first implemented the Slave Review Act and emancipated the many slaves illegally held by powerful clans.*" (1973)

Discriminative. At the Postwar pole, *haebang* names the event directly as a chapter heading: "*Our country's liberation*" (우리 나라의 해방, 1954). At the Authoritarian pole, the word appears within cultural-history narratives rather than as a standalone political category.

Alignment residual. Procrustes reverses cPCA, placing slave emancipation at the negative pole and national liberation at the positive. This reversal confirms that the two senses of the word occupy distinct positions in the embedding space, and the axes rotate through that space differently.

Authoritarian → Democratic. *Contrastive variance.* The polysemy structure persists but rotates. At the negative pole (Authoritarian direction), the word again means premodern emancipation:

> 공민왕은 신돈을 등용하여, 권문세족이 빼앗은 토지를 원래의 주인에게 돌려주고 불법적으로 소유한 노비를 해방시켰다.
>
> "*King Kongmin appointed Sindon, returned land seized by the powerful families to its original owners, and emancipated slaves held illegally.*" (2006)

At the positive pole (Democratic direction), the word enters a social-movement register absent from authoritarian textbooks:

> 여성 운동 단체들은 여성의 사회적 지위 향상과 여성 해방을 위한 사회 운동을 전개하였다.
>
> "*Women's movement organizations carried out social movements for the improvement of women's social status and women's liberation.*" (2016)

Discriminative. The logistic axis captures a different contrast. At the Authoritarian pole, *haebang* means 1945 national liberation:

> 민족은 해방되었으나 고대하던 독립이 즉시 이루어지지는 않았다.
>
> "*The nation was liberated, but the long-awaited independence was not immediately achieved.*" (1973)

At the Democratic pole, it appears within the nationalist-socialist split:

> 사회주의 사상이 유입되고 그 세력이 급속히 확산되면서 독립 운동계는 민족 해방을 부르짖는 민족주의 계열과 계급 해방을 목표로 하는 사회주의 세력으로 양분되었다.
>
> "*As socialist thought flowed in and its influence spread rapidly, the independence movement was divided into a nationalist faction calling for national liberation and a socialist faction aiming for class liberation.*" (2007)

Across both transitions, the contrastive axes for *haebang* separate senses of the word rather than political framings of the same event. The modern Korean sense of the term, liberation from colonial rule, appears at similar positions regardless of period. What shifts is the word's secondary usage: premodern slave emancipation in authoritarian textbooks gives way to social-movement liberation (women's, class) in democratic ones.

G.5 사회 (*sahoe*, society)

Postwar → Authoritarian. *Contrastive variance.* At the Postwar-direction pole, *sahoe* appears as a chapter-heading label:

> 근대 사회의 성장
>
> "*Growth of Modern Society.*" (1987)

At the Authoritarian-direction pole, the term appears in a concrete description of premodern social hierarchy:

> 따라서, 모든 정치는 왕과 귀족이 주도하였고, 신라의 골품 제도와 같은 엄격한 신분 제도에 따라 사회가 유지되었다.
>
> "*Accordingly, all politics were led by kings and aristocrats, and society was maintained according to strict status systems such as Silla's bone-rank system.*" (1987)

Discriminative. The logistic axis captures a completely different register. At the Postwar pole, *sahoe* appears not in Korean social history but in a description of the United Nations:

> 사무 총장은, 안전 보장 이사회에서 사람을 골라서, 총회에서 반수 이상의 추천을 얻어야 한다.
>
> "*The Secretary-General must be selected by the Security Council and receive the recommendation of more than half the General Assembly.*" (1954)

At the Authoritarian pole, the term appears in Confucian moral language:

> 조선 사회에서는 유교적인 가족 질서를 바탕으로 충효의 윤리가 강조되었다.
>
> "*In Chosŏn society, the ethics of loyalty and filial piety were emphasized on the basis of Confucian family order.*" (1987)

The logistic axis thus identifies the term's postwar usage in international-institutional contexts as discriminative, a register that disappears under authoritarianism.

Alignment residual. Procrustes agrees with cPCA at the negative pole (same chapter-heading sentence). At the positive pole, Procrustes places a shorter Confucian-society label:

> 유교 중심의 양반 사회
>
> "*Confucian-centered yangban society.*" (1973)

Authoritarian → Democratic. *Contrastive variance.* The negative pole (Authoritarian direction) brings the nationalist-socialist alliance into view:

> 한편, 단일화된 민족 운동을 위한 노력이 계속 추진되어 민족주의계와 사회주의계가 합작하여 신간회를 조직하였다.
>
> "*Meanwhile, efforts toward a unified national movement continued, and the nationalist and socialist factions jointly organized the Singanhoe.*" (1981)

At the positive pole (Democratic direction):

> 옛날보다 지금의 사회 제도는 더 복잡해졌다.
>
> "*Compared to the past, today's social institutions have become more complex.*" (1973)

Alignment residual. Procrustes reverses cPCA for this transition. The Singanhoe sentence appears at the Procrustes *positive* pole, and the "more complex" sentence at the Procrustes *negative* pole. The two axes identify the same semantic contrast but orient it in opposite directions, indicating that the dimension of maximum variance difference and the dimension of maximum structural reorganization are not aligned for *sahoe* in this transition.

G.6 통일 (*t'ongil*, unification)

Postwar → Authoritarian. *Contrastive variance.* At the Postwar-direction pole, *t'ongil* refers to pre-modern historical unification:

> 3층 석탑(석가탑) 통일 신라 시대의 대표적인 3층 석탑이다.
>
> "*Three-story stone pagoda* (*Sŏkkat'ap*): *A representative three-story stone pagoda of the Unified Silla period.*" (1981)

The Authoritarian-direction pole is the National Education Charter:

> 길이 후손에 물려줄 영광된 통일 조국의 앞날을 내다보며, 신념과 긍지를 지닌 근면한 국민으로서, 민족의 슬기를 모아 줄기찬 노력으로, 새 역사를 창조하자.
>
> "*Looking ahead to the future of a glorious unified fatherland to be bequeathed to posterity, let us, as diligent citizens with conviction and pride, gather our nation's wisdom and create a new history through persistent effort.*" (1991)

Discriminative. The logistic axis separates two ways of referencing Unified Silla. At the Postwar pole, it appears within evaluative cultural claims:

> 삼국 시대와 통일 신라 시대의 찬란한 불교 예술이 그것이며, 고려 시대의 자기와 활자, 조선 시대의 측우기나 한글 같은 것은 어느 나라에 비할 바가 없이 훌륭한 것이다.
>
> "*The splendid Buddhist art of the Three Kingdoms and Unified Silla periods, the celadon and movable type of the Koryŏ period, and the rain gauge and Hangŭl of the Chosŏn period are all magnificent beyond comparison with any other country.*" (1963)

At the Authoritarian pole, it appears in descriptive captions:

> 포석정 통일 신라 시대의 건축물로, 돌로 도랑 모양을 만들어 물이 흐르게 하였다.
>
> "*P'osŏkchŏng*: *A Unified Silla structure in which channels were carved into stone to let water flow.*" (1987)

Alignment residual. Procrustes reverses cPCA. The Charter appears at the *negative* pole, while Buddhist material culture moves to the positive pole. As with other terms, the reversal indicates that variance expansion and structural repositioning operate along different geometric axes.

Authoritarian → Democratic. *Contrastive variance.* The Authoritarian-direction pole confines *t'ongil* to historical reference:

> 통일 신라의 왕권 강화
> "*Strengthening of royal authority in Unified Silla.*" (1987)

The Democratic-direction pole is contemporary policy:

> 통일 정책과 남북 대화
> "*Unification policy and inter-Korean dialogue.*" (2007)

The highest-scoring pole exemplar for cPCA is already quoted in the body of the paper.

Discriminative. The logistic axis brings in a global comparative register absent from both other axes:

> 특히, 낫세르는 수에즈 운하의 국유화를 단행하여 민족의 의지를 보였고, 시리아와 함께 통일 아랍 공화국을 형성하기도 하였다.
> "*In particular, Nasser demonstrated the will of the nation by nationalizing the Suez Canal, and also formed the United Arab Republic with Syria.*" (1973)

This sentence, which uses *t'ongil* in a non-Korean context, is the most discriminative of the Authoritarian period. The logistic classifier treats comparative world-history references to unification as distinctly authoritarian-era language.

Alignment residual. Procrustes again reverses cPCA, placing "Unification policy and inter-Korean dialogue" at the negative pole and "Unified Silla strengthened royal authority" at the positive.

G.7 역사 (*yŏksa*, history)

Postwar → Authoritarian. *Contrastive variance.* The Postwar-direction pole is the National Education Charter, where *yŏksa* is part of the injunction to create new history:

> 길이 후손에 몸려준 영광된 통일 조국의 앞날을 내다보며, 신념과 공지를 지닌 근면한 국민으로서, 민족의 솔기를 모아 줄기찬 노력으로, 새 역사를 창조하자.
> "*Looking ahead to the future of a glorious unified fatherland to be bequeathed to posterity, let us, as diligent citizens with conviction and pride, gather our nation's wisdom and create a new history through persistent effort.*" (1990)

At the Authoritarian-direction pole, the term is in a more concrete pedagogical register:

> 역사를 알아보는 데 이용되는 자료에는 여러 가지가 있다.
>
> "*There are many kinds of sources used for investigating history.*" (1987)

Discriminative. The logistic axis identifies a different contrast. At the Postwar pole, *yŏksa* appears within a discussion of historiographic persecution:

> 사필로 말미암은 사림의 화인 까닭에 역사가는 이를 특히 사화라 이른다.
>
> "*Because the literati suffered calamity on account of the historian's brush, historians call these events specifically 'literati purges.'*"

At the Authoritarian pole, history becomes a programmatic declaration:

> 역사란 지나가 버린 과거의 사실과 연대를 공부하고 기억하는 학문이 아니다.
>
> "*History is not a discipline of studying and memorizing the facts and dates of a vanished past.*" (1973)

Alignment residual. Procrustes fully agrees with cPCA on this transition: the Charter at the negative pole, pedagogical sources at the positive.

Authoritarian → Democratic. *Contrastive variance.* At the negative pole (Authoritarian direction), democratic textbooks take a comparative and reflexive approach:

> 올바른 역사 의식을 가지고 우리 역사를 객관적으로 이해하려고 하면, 우리 역사뿐만 아니라 주변 민족, 주변 사회의 역사도 바르게 이해해야 한다.
>
> "*To understand our history objectively with a correct historical consciousness, we must properly understand not only our history but also the history of neighboring nations and societies.*" (2004)

Discriminative. The logistic axis captures a different dimension. At the Authoritarian pole, *yŏksa* appears in the narrative of World War II:

> 이로써 인류 역사상 최대의 전쟁인 제2차 세계 대전이 연합국의 승리로 끝났다.
>
> "*Thus the largest war in human history, the Second World War, ended with the Allied victory.*" (1973)

Alignment residual. The Procrustes negative pole captures a similar comparative framing to cPCA but with different emphasis:

따라서 역사를 바르게 이해한다는 것은 세계사적 보편성과 지역적 특수성을 균형 있게 파악함을 의미하는 것이다.

"*Therefore, understanding history correctly means grasping the balance between world-historical universality and regional particularity.*" (2002)

G.8 국가 (*kukka*, state)

Postwar → Authoritarian. *Contrastive variance.* The Postwar-direction pole places *kukka* in a fiscal register:

국가 재정과 농민의 부담

"*State finances and the burden on farmers.*" (1987)

At the Authoritarian-direction pole, the term is part of narratives of ancient state formation:

삼국은 점차 왕을 중심으로 하는 중앙 집권적인 국가로 발전하였다.

"*The Three Kingdoms gradually developed into centralized states centered on the king.*" (1981)

Discriminative. The logistic axis captures a Cold War ideological dimension. At the Postwar pole:

만주주의 국가의 국회와 공산주의 국가의 국회는 어떻게 다른지 비교하여 보자.

"*Let us compare how the parliament of a democratic state and the parliament of a communist state differ.*" (1963)

This comparative Cold War framing of *kukka* is the most discriminative sentence for the Postwar period, a register that disappears from authoritarian-era textbooks, where ideological comparison gives way to historical narration.

Alignment residual. Procrustes agrees with cPCA on this transition: the same sentences occupy the same poles.

Authoritarian → Democratic. *Contrastive variance.* At the negative pole (Authoritarian direction), democratic textbooks extend *kukka* into Cold War geopolitics:

> 제24회 서울 올림픽 대회 개막 행사(1988)　동서 냉전의 갈등으로 한동안 올림픽에 참여 하지 않았던 공산권 국가들까지 참가하여 동서 화합의 축제가 되었다.
>
> "*Opening ceremony of the 24th Seoul Olympic Games (1988): Even communist-bloc states that had not participated in the Olympics for some time due to East-West Cold War tensions joined, making it a festival of East-West reconciliation.*" (2002)

At the positive pole (Democratic direction), *kukka* is a premodern institutional descriptor:

> 통일 신라는 국가의 수입을 늘리고 귀족 세력을 억제하기 위하여 토지 제도를 개편하였다.
>
> "*Unified Silla reformed its land system to increase state revenue and restrain aristocratic power.*" (1995)

Discriminative. The logistic axis picks up a different sentence at the Authoritarian pole, a declaration of democratic identity:

> 우리 나라에는 종교가 여러 가지 있으며 국교가 없이 신앙의 자유가 보장되어 있는 것은, 우리 나라가 민주주의 국가임을 말해 주는 것이다.
>
> "*The fact that our country has many religions and guarantees freedom of belief without a state religion shows that our country is a democratic state.*" (1973)

Alignment residual. At the Procrustes positive pole, a chronological timeline from 1989 marks the diplomatic opening to Eastern Europe:

> 1987: 6.29 민주화 선언　1988: 노태우 정부 성립, 제24회 서울 올림픽 대회　1989: 헝가리, 폴란드 등 동구권 국가와 수교
>
> "*1987: June 29 Democratization Declaration. 1988: Inauguration of the Roh Tae-woo government, 24th Seoul Olympic Games. 1989: Establishment of diplomatic relations with Hungary, Poland, and other Eastern bloc states.*" (2001)

At the Procrustes negative pole, *kukka* appears in the same 15th-century fiscal context as the cPCA positive pole, reversing the orientation.

Additional exemplar sentences for other terms are available in the replication archive.

Appendix H: Full-text robustness and boundary terms

The main text describes the term-discovery pipeline as operating on `full_text` windows. An earlier version of the pipeline fed the three EXAONE models a noun-filtered `processed_text` stream derived from KiwiPiePy NNG/NNP tokenization. That choice worked cleanly for frequency analysis but left a methodological question open: would the same twelve terms survive if the models read natural prose instead of a decontextualized noun bag? This appendix reports the robustness rerun that answers that question, and it also describes a small set of terms that surfaced in a single model under full-text extraction without meeting cross-model consensus.

H.1 Full-text vs. processed-text consensus

We re-ran the complete three-model pipeline (EXAONE 2.4B, 7.8B, and 32B-q4_K_M) on `full_text` windows with the same adaptive sampling, extraction prompt, and scoring composite used for the `processed_text` baseline. Table S7 reports the resulting cross-model scores alongside the earlier `processed_text` averages for each of the twelve final terms.

Term	Models	Avg freq (FT)	Avg freq (PT)	Retention basis
liberation (*haebang*, 해방)	2/3	837.9	541.8	Consensus (FT)
politics (*chŏngch'i*, 정치)	2/3	824.7	854.1	Consensus (FT)
nation (*minjok*, 민족)	3/3	785.0	835.0	Consensus (FT)
territory (*yŏngt'o*, 영토)	2/3	771.5	488.7	Consensus (FT)
unification (*t'ongil*, 통일)	3/3	664.3	645.1	Consensus (FT)
democracy (*minjujuŭi*, 민주주의)	3/3	641.1	568.1	Consensus (FT)
history (*yŏksa*, 역사)	3/3	640.2	597.8	Consensus (FT)
state (*kukka*, 국가)	3/3	635.4	619.9	Consensus (FT)
independence (*tongnip*, 독립)	3/3	626.0	518.0	Consensus (FT)
community (*kongdongch'e*, 공동체)	2/3	557.0	450.1	Consensus (FT)
citizen (*kungmin*, 국민)	3/3	367.9	423.1	Consensus (FT)
society (*sahoe*, 사회)	2/3	186.2	328.3	Theoretical retention (PT)

Table S7: Full-text (FT) vs. processed-text (PT) cross-model consensus for the final twelve terms. "Models" reports how many of the three EXAONE models selected the term above their per-model threshold under full-text extraction. Avg freq values are cross-model means of the weighted score described in the main text. Consensus threshold: selected by $\geq$2 models with cross-model avg freq $\geq$300.

Eleven of the twelve terms pass the full-text consensus filter directly, with an overall Jaccard similarity of 0.917 against the processed-text selection. Four terms score higher under full-text than under processed-text extraction, most notably *haebang* (liberation, 837.9 vs. 541.8), *yŏngt'o* (territory, 771.5 vs. 488.7), and *kongdongch'e* (community, 557.0 vs. 450.1). These gains are modest in substantive terms but they reduce a prior concern. *Kongdongch'e* had been borderline in the processed-text run, and its comfortable consensus under full-text extraction suggests it is a genuinely recurrent identity concept rather than an artifact of noun filtering.

The single exception is *sahoe* (사회, society). It is selected by two of the three EXAONE models under full-text extraction, which clears the cross-model threshold, but its cross-model mean weighted frequency is 186.2, well below the 300 cutoff. Under processed-text extraction it scored 328.3 and met both filters cleanly. The pattern is consistent with what one would expect from a word that is semantically central but grammatically diffuse: noun

filtering removes surrounding function words and concentrates *sahoe*'s co-occurrence with other identity nouns, raising its weighted score, whereas full-text windows dilute the same contexts with particles and inflected forms.

We retain *sahoe* in the final twelve-term set for three reasons. First, it passes cross-model consensus under full-text and it passes both filters under processed-text, so the retention is consistent with at least one complete run of the discovery pipeline. Second, both topic-modeling robustness checks corroborate it: BERTopic places *sahoe* as the top representative word of one topic and within the top thirty of several others (Appendix A, Table S2), and LDA recovers it with topic-level rank 1 in multiple topics. Third, qualitative inspection locates *sahoe* within the civic-societal cluster of identity vocabulary rather than on its periphery. The retention is flagged in `final_selected_terms.csv` via a `retention_basis` column valued `theoretical_retention_processed_text`, and the cross-model row in Table S7 preserves the underlying scores for transparency.

H.2 Boundary terms under full-text extraction

A small number of candidate terms surfaced in a single model under full-text extraction without meeting cross-model consensus. Table S8 reports the three most prominent: *Han'guk* (한국, Korea), *uri* (우리, we/our), and *Han'gugin* (한국인, Korean person). Each appears in the EXAONE-Deep 32B selections at high weighted scores but is absent from both smaller models' selections.

Term	2.4B	7.8B	32B	Corpus freq.
Korea (*Han'guk*, 한국)	—	—	1,215.4	6,173
we/our (*uri*, 우리)	—	—	1,134.6	14,590
Korean person (*Han'gugin*, 한국인)	—	—	825.1	908

Table S8: 32B-only full-text boundary terms. Columns 2.4B, 7.8B, and 32B report the weighted score when a model selected the term above its per-model threshold; a dash indicates that the model did not select the term at all. "Corpus freq." reports the raw occurrence count in the merged 67-textbook corpus. None of these terms meets the cross-model consensus requirement of selection by $\geq$2 models.

The three boundary terms have different profiles. *Han'guk* is a high-frequency proper noun that appears in 56 of 67 textbooks and names the country directly. *Uri* is the first-person plural possessive pronoun and the most common token in the corpus, with 14,590 occurrences across all 67 textbooks; it is the standard collective marker in Korean prose rather than an identity category per se. *Han'gugin* is lower-frequency and more concentrated, appearing in 39 textbooks; it is the explicit noun for "Korean person" and therefore a more substantive candidate than the other two.

Topic-modeling artifacts provide partial corroboration but not consistent clustering. BERTopic places *Han'guk* in the top-thirty representations of ten topics and *Han'gugin* in six, typically colonial-period topics centered on diaspora, policing, or education rather than on the core identity cluster that surfaces the twelve selected terms. *Uri* does not appear in the top-thirty representation of any BERTopic topic, because c-TF-IDF down-weights tokens that are distributed broadly across the corpus. LDA shows a similar pattern: *Han'guk* appears in two topics' top-ten representations but is not among the top words in any topic dedicated to identity or political discourse. This is consistent with the interpretation that *Han'guk* and *uri* index corpus-wide ubiquity rather than a distinct identity concept, and that *Han'gugin*, while more specific, does not meet the evidentiary bar required by cross-model consensus.

The failure of these terms to clear the consensus filter is informative. It illustrates what

the filter is designed to absorb. Generative extraction from a single large model is probabilistic and sensitive to prompt-specific anchoring; a 32B model may surface terms that a smaller model in the same family does not, and vice versa. Requiring agreement across at least two of three independent model runs reduces the influence of any single model's idiosyncratic choices. We flag *Han'gugin* in particular as a reasonable candidate for future work on explicit ethnic-national nouns, especially in comparisons between colonial-era and post-liberation usages, but it is not part of the twelve-term set analyzed in the main paper.